Cryptography and Cybersecurity

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Homework 3:

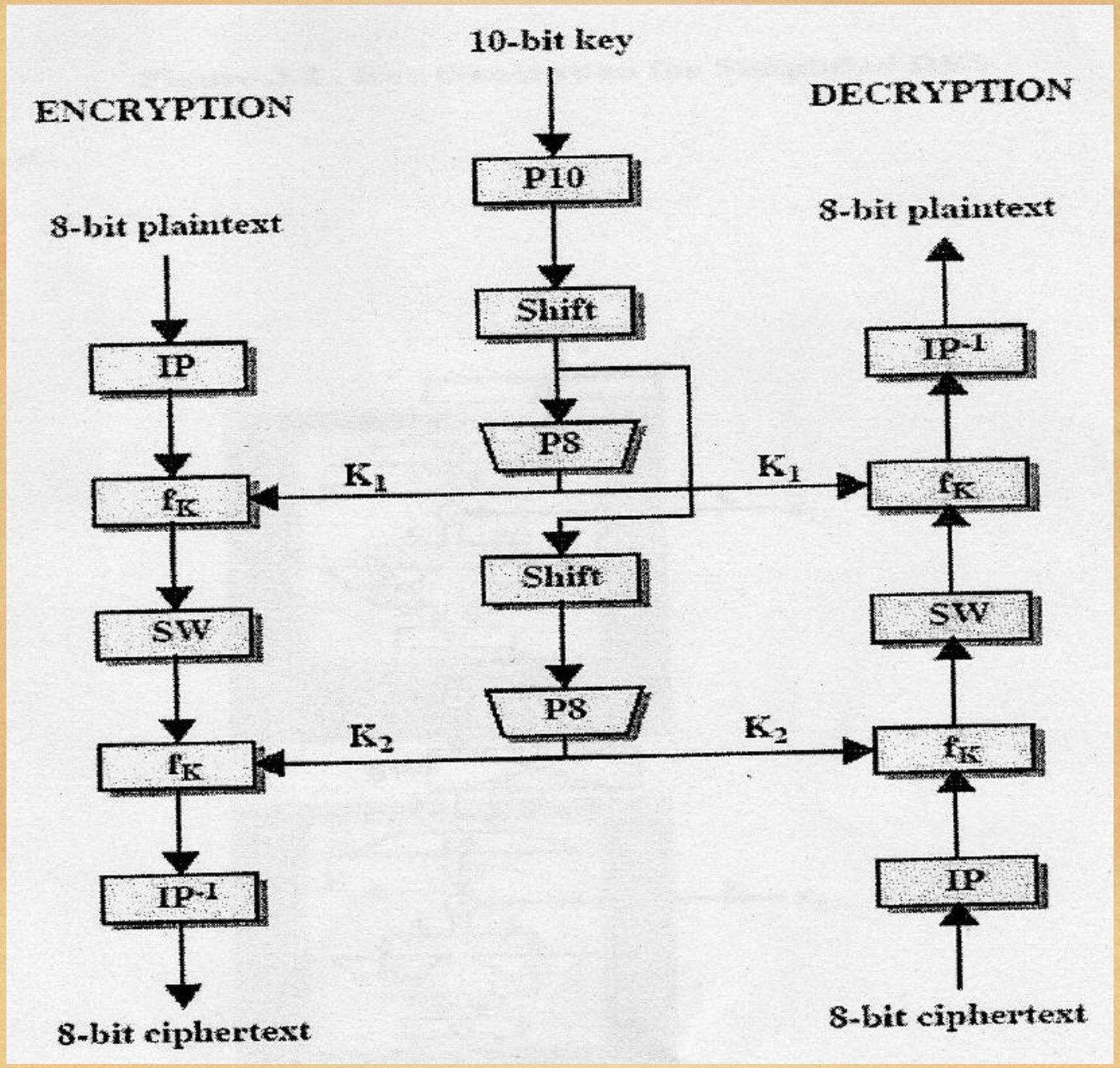
Implementation of a Symmetric Cipher

Implementation of a Symmetric Cipher

The symmetric cipher in this homework is a simplified version of the Data Encryption Standard (S-DES).
Using the MATLAB environment, implement the encryption and decryption modules of the S-DES algorithm.
The program should be able to:

1. Receive a message in the form of a string of alphanumeric characters as input, say M, and convert it into a string of bits, say P.
2. Encrypt the plaintext P, producing the corresponding ciphertext C
3. Decrypt the ciphertext C, thereby obtaining the plaintext P
4. Convert the plaintext P into the original message M

Check the operation of your implementation by testing different messages, verifying that the algorithm is able to output the correct message after decryption.


10-bit key
ENCRYPTION
DECRYPTION
P10
8-bit plaintext
8-bit plaintext
Shift
IP
IP-1
P8
K1
K1
fK
fK
Shift
SW
SW
P8
K2
K2
fK
fK
IP-1
IP
8-bit ciphertext
8-bit ciphertext

Specifications of S-DES

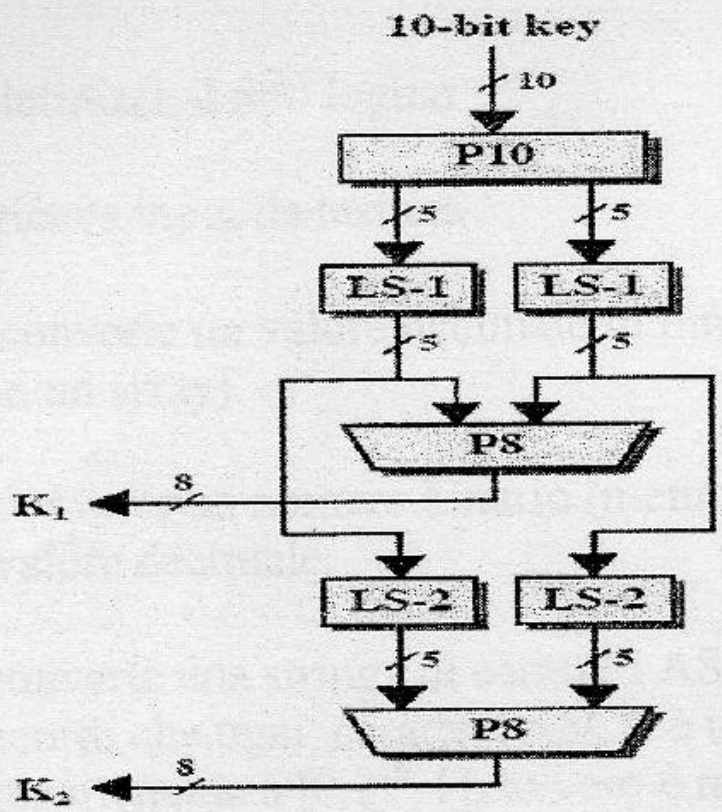


Figure 3.2 Key Generation for Simplified DES

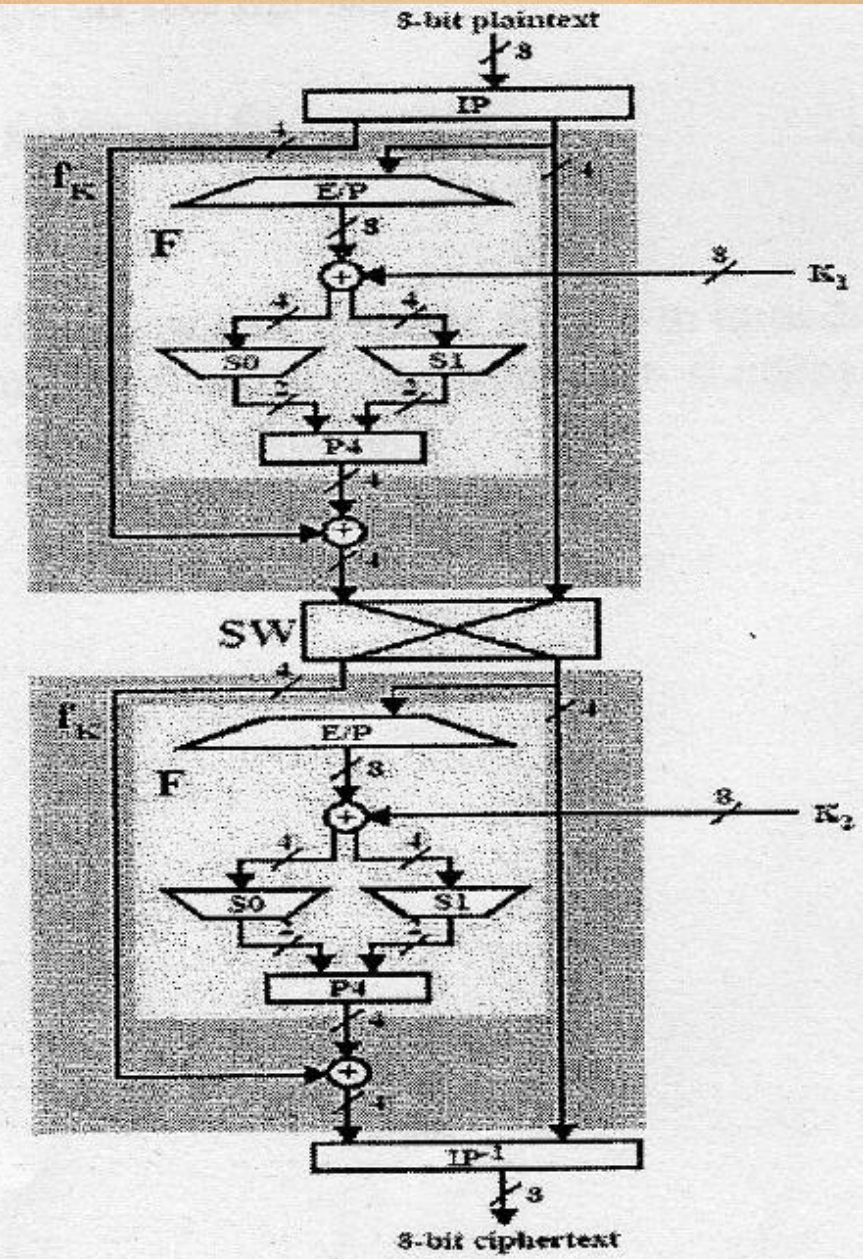


Figure 3.3 Simplified DES Encryption Detail

Specifications of S-DES

1. $IP = [2, 6, 3, 1, 4, 8, 5, 7]$; $IP^{-1} = [4, 1, 3, 5, 7, 2, 8, 6]$
2. $P10 = [3, 5, 2, 7, 4, 10, 1, 9, 8, 6]$; $P8 = [6, 3, 7, 4, 8, 5, 10, 9]$
3. $LS - n =$ n-bit left circular shift
4. $E/P = [4, 1, 2, 3, 2, 3, 4, 1]$; $P4 = [2, 4, 3, 1]$
5. SW = Exchanges left and right halves of input data

$$S0 = \begin{bmatrix} 1 & 0 & 3 & 2 \\ 3 & 2 & 1 & 0 \\ 0 & 2 & 1 & 3 \\ 3 & 1 & 3 & 2 \end{bmatrix}, \qquad S1 = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 2 & 0 & 1 & 3 \\ 3 & 0 & 1 & 0 \\ 2 & 1 & 0 & 3 \end{bmatrix}$$